Technical Note: Snake Game Architecture & Specifications

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1. System Overview

This technical note details the structural parameters, runtime mechanics, and automated verification harness of the retro Snake game implemented in pure HTML5 and Canvas 2D. The architecture decouples visual rendering from state updates using a fixed-frequency tick engine and a non-reversing input queue.

2. Visual Architecture & Gameplay Capture

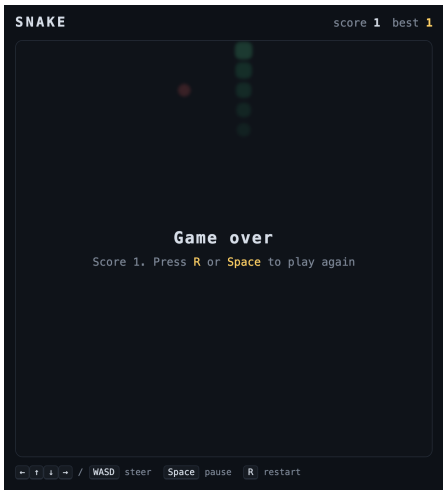


Figure 1: Active gameplay state captured over Chrome DevTools Protocol (CDP) showing score HUD, arena border, snake body segments, overlay prompt, and controls.

3. Dimensional Specifications & Engine Constants

Parameter	Quantitative Value	Architectural Description & Rationale
Arena Grid	21×21 discrete cells	Odd-numbered dimensions allowing an exact symmetrical center origin.
Canvas Resolution	500×500 px	Coordinate raster scaled responsively via CSS `min(90vmin, 560px)`.
Unit Cell Pitch	23.8×23.8 px	Effective grid cell dimension on the rendering canvas.
Tick Interval	110 ms (9.09 Hz)	Fixed simulation step balancing playability and collision responsiveness.
Initial Snake Length	4 body segments	Starting configuration instantiated horizontally at row center ($y = 10$).
Input Turn Queue	Depth: 3 entries	FIFO buffer preventing key jamming and filtering out 180° self-reversals.
Food Entity	1 unit cell (20 px)	Spawned in unallocated free cells using deterministic 32-bit PRNG.

4. Commands to Rebuild, Execute & Test

- **Direct Interactive Viewer** (open in local browser):
```
open snake/index.html
```
- **Rebuild Presentation Slides & PDF Note**:
```
quarto render slides-with-animation-pdf-note/slides.qmd
tectonic slides-with-animation-pdf-note/note.tex
```
- **Automated Headless CDP Verification Test**:
```
uv run --with websocket-client python /tmp/drive_snake.py
```

5. Project Links

- **Interactive Viewer**: github.com/kafcoo/mlr-agent-lab/blob/main/snake/index.html
- **Automation Script**: github.com/kafcoo/mlr-agent-lab (drive_snake.py)